Lending Your Way to Currency Dominance: Chinese Development Finance and RMB Trade Invoicing

Abstract: This paper examines whether Chinese overseas lending promotes the use the renminbi (RMB) as an invoicing currency. Combining detailed loan-level data from AidData with bilateral trade data, we find that higher Chinese loan-to-GDP exposure significantly strengthens RMB exchange rate pass-through into recipient countries' import prices and quantities from China while weakening dollar exchange rate pass-through. The effects are concentrated in loans extended by Chinese policy banks, non-resource-backed loans, and infrastructure-related products. These patterns are consistent with a procurement-tying mechanism whereby Chinese official loans raise demand for related Chinese goods that are more likely to be invoiced in RMB. We rule out alternative explanations based on bilateral swap lines, borrower financial distress, political alignment, and currency matching. We also conduct placebo tests using World Bank loans and Japanese and U.S. official development assistance further confirm that the loan effects are unique to Chinese official lending.

1. Introduction

The internationalization of the renminbi (RMB) has been a central objective of Chinese economic policy since the 2008 global financial crisis. Chinese government has implemented a range of initiatives to expand the currency's international role, including the promotion of cross-border RMB trade settlement, the development of offshore RMB markets, the issuance of RMB-denominated bonds in Hong Kong, and the establishment of bilateral currency swap lines with foreign central banks. Although the effects of these policies have been examined in the literature, the potential role of Chinese overseas lending to RMB invoicing remains unexplored.

Over the past two decades, China has increased its overseas lending from $2.24 billion in 2000 to $99.42 billion in 2023, both in nominal US dollars. Chinese oversea lending is now comparable in magnitude to that of the largest multilateral lender World Bank, although it remains below the combined lending of IBRD and IDA (Figure 1). China's lending footprint in the developing world currently rivals that of the World Bank (Horn et al. 2021, Horn et al. 2025). Through its policy banks and other official lenders, China has extended hundreds of billions of dollars in loans to both developing and advanced economies. In this paper, we examine whether Chinese state-directed lending creates the commercial infrastructure for currency internationalization, even in the absence of the deep financial markets and safe assets that theory identifies as prerequisites.

Understanding the link between Chinse overseas lending and RMB invoicing is important for two reasons. First, it sheds light on a potential lending-to-invoicing channel that may accelerate RMB internationalization. Second, it helps explain how China's rise a s global creditor may influence the currency choice in bilateral trade with China. While a growing literature has studied the determinants of invoicing currency choice and the economic consequences of Chinese overseas lending, the connection between the two remains largely unexplored.

We combine detailed Chinese overseas lending data from AidData with recipient countries' bilateral import data from UN Comtrade. We construct a country-year

measure of Chinese official loan exposure scaled by recipient GDP and interact this variable with the recipient's bilateral exchange rates against the RMB and dollar in standard exchange rate pass-through specifications for both import prices and quantities. We find that greater Chinese loan exposure significantly strengthens the sensitivity of import prices and quantities to the RMB exchange rate, while weakening the sensitivity to the dollar exchange rate. We interpret the stronger RMB exchange rate pass-through as evidence of increased RMB invoicing in recipient countries' imports from China.

The main mechanism is that Chinese overseas loans are predominantly directed toward infrastructure projects. This raises recipient countries' imports of infrastructure-related products from China, which are more likely to be invoiced in RMB. Consistent with this channel, the effects are stronger for loans extended by Chinese policy banks, for non-resource-backed loans, and for infrastructure-related products.

Our results can rule out several alternative explanations. The results are not driven by the presence of bilateral RMB swap lines or RMB clearing banks, by borrower's financial distress with China, by political alignment with China, or by currency matching between the original denomination of the loans and trade invoicing. Our findings are further supported by a series of placebo tests using World Bank loan and Japanese and USA ODA, indicating that the loan effects are specific to Chinese overseas lending.

Our paper relates to three strands of literature. First, we contribute to the growing literature on Chinese overseas lending. A rapidly expanding body of work has documented China's emergence as the world's largest official creditor. Horn, Reinhart, and Trebesch (2021) provide comprehensive description of the scale, structure and geographic distribution of Chinese official loans. Horn, Reinhart, and Trebesch (2025) focus specifically on lending to developing countries, while Jiang (2025) examines the effects of Chinese lending on recipients' financial cycles, including exchange rates, equity prices, bond yields, and capital flows. These studies have greatly improved our understanding of the volume and macroeconomic consequences of Chinese overseas finance. However, they have largely left unexplored the implications of Chinese lending

for the currency choice of trade between recipient countries and China.

Second, our paper relates to the extensive literature on the dominant role of the US dollar in international trade invoicing. A large body of empirical work documents that the dollar remains the predominant invoicing currency even in trade that does not invole the US (Gopinath, 2015; Gopinath et al., 2010, 2020; Boz et al., 2022; Goldberg and Tille, 2009, Giuliano and Luttini, 2020). Consistent with this literature, we find that the dollar continues to play an important role in recipient countries' imports from China. At the same time, our results show that the RMB also exerts a statistically significant influence on bilateral exchange rate pass-through, indicating that RMB invoicing is becoming increasingly relevant in China and recipient country's bilateral trade.

Third, we contribute to the literature on the determinants of invoicing currency choice. Existing research has identified macroeconomic force such as demand for safe assets and network externalities (Gopinath et al., 2021; Farhi and Maggiori, 2018; Eichengreen et al., 2017) as well as microeconomic factors including firm size, imported input intensity, competitor's currency choices, import and export market shares, and product differentiation (Amiti et al., 2014, 2022; Goldberg and Tille, 2016). More recently, Benguria and Novy (2026) and Bahaj and Reis (2026) examine the role of RMB currency swap lines in promoting RMB usage. We complement these studies by proposing a new lending-to-invoicing channel, Chinese overseas lending creates real-side procurement linkages, particularly through infrastructure projects financed by Chinese stated-owned policy banks, which increases the RMB exchange rate pass-through in trade.

[Figure 1 is about here]

2. Data

Our empirical analysis draws on two main datasets. The first is AidData's China's Global Loans and Grants Dataset, which records Chinese overseas lending and is used to construct our independent variable. The second is UN Comtrade, which records recipient countries' imports from China and is used to construct our dependent variables.

We also introduce a set of macroeconomic control variables.

2.1 Chinese Overseas Lending

We obtain data on Chinese overseas lending from AidData's China's Global Loans and Grants Dataset, Version 1.0[1]. This comprehensive dataset covers 33580 projects and activities across 217 low-, middle, and high-income countries, supported by grants and loans from Chinese official sector institutions totaling nearly $2.2 trillion. We focus exclusively on loans (flow type as "Loan"). For each project the dataset provides the recipient country, commitment year, financial value (nominal and constant U.S. dollars), project purpose, original currency of denomination (e.g., RMB, U.S. dollar, British pound, etc.), and the type of funding agency (e.g., Chinese government agency, Chinese stated-owned policy bank, Chinese stated-owned commercial bank, etc.). This granularity allows us to examine loan heterogeneity in subsequent mechanism analyses.

Following Dreher et al. (2021), we refine the sample as follows. We exclude projects identified only at the regional level (e.g., Africa or Asia) rather than for a specific country, as well as multi-country projects. We also remove projects that were double-counted or subsequently canceled. We also drop projects in which the jurisdiction of direct receiving agency is China itself and retain only projects whose information sources include at least one official source. After applying these restrictions, the final sample consists of 9785 lending projects over 2000-2023, covering both developing and developed countries.

We construct our independent variable as the ratio of annual nominal loan commitments to the recipient country's nominal GDP, scaled by 100 (loan-to-GDP ratio). Figure 2 plots the global distribution of Chinese overseas lending over 2000-2023. Panel A shows each country's mean annual loan-to-GDP ratio[2]; Panel B shows cumulative value in constant 2023 U.S. dollars. Darker shades indicate higher exposure

[1] Retrieved from http://www.aiddata.org/data/chinas-global-loans-and-grants-dataset-1-0.

[2] Marshall Islands records a mean loan-to GDP ratio exceeding 100 percent but is excluded from the regression sample because its currency is pegged to the US dollar. We therefore do not need to worry about this outlier.

or larger cumulative values.

[Figure 2 is about here]

2.2 UN Comtrade Data

Our main dependent variable is the log change in the unit value (price) of recipient countries' imports from China at the country-product-year level, where products are defined at the HS 2002 six-digit level. We obtain the data from the UN Comtrade database. Unit values are calculated by dividing observed import values by quantities and serve as a proxy for prices:

$$\Delta p_{iht} \equiv \Delta \log\left(\frac{import\ value_{iht}}{import\ quantity_{iht}}\right)$$

Import values are converted into the recipient country's currency. Following Li et al. (2015), we drop observations subject to noise or measurement error: (1) products with inconsistent units or missing quantity information, and (2) special categories such as arms (HS 93), antiques (HS97), and other special codes (HS98-HS99). For consistency across HS revisions, we apply publicly available concordances that map HS 2007, 2012, 2017, and 2022 codes to the HS 2002 system. To mitigate the influence of outliers we winsorize the annual growth rates of unit values and quantities at the top and bottom 1 percent of the distribution by year.

2.3 Other Control Variables and Sample Construction

Exchange rates are taken from the World Bank's "alternative conversion factor" series, which corrects for redenominations and currency substitutions and is measured as annual averages of daily rates. Following Gopinath et al. (2020) and Boz et al. (2022), we control for China's CPI inflation (to capture exporter's production costs) and recipient countries' real GDP growth rate (to capture import demand). These series are drawn from the World Development Indicators. Further details on our data are provided in online Appendix A.

We exclude dollarized countries and developed countries from the main sample. Following Gopinath et al. (2020), we classify a country as dollar-pegged if its official

currency is the U.S. dollar or its bilateral exchange rate is strictly fixed against the dollar. Developing countries are defined as those eligible for official development assistance according to AidData (primarily low-, lower-middle-, and upper-middle-income countries by OECD classification). The detailed country list is in Appendix A. The final sample comprises 116 countries. Because the RMB was pegged to the U.S. dollar until 2005, we restrict the sample period to 2006-2023. We focus on non-commodity goods, defining commodities broadly as HS chapters 1-27 and 72-83 (animal, vegetable, food, mineral, and metal products), following Gopinath et al. (2020). Appendix Table A1 reports the summary statistics of the main variables.

3. Empirical Strategy and Baseline Results

3.1 Empirical Specification

We begin with standard exchange rate pass-through regression that relate changes in recipient countries' import prices from China to changes in nominal exchange rates. Following Burstein and Gopinath (2014) and augmenting the specification with the dollar exchange rate, we estimate:

$$\Delta p_{iht} = \beta_1 \Delta e_{it}^{RMB} + \beta_2 \Delta e_{it}^{\$} + \theta_1 \Delta CPI_t^{China} + \theta_2 \Delta RGDP_{i,t} + \varphi_{ih} + \epsilon_{iht} \quad (1)$$

where i, h, and t index recipient country, product (HS2002, six-digit) and year, respectively. Δp_{iht} is the log change in the import price of product h from China, expressed in the recipient country's currency. Δe_{it}^{RMB} is the log change in the bilateral nominal exchange rate (local currency units per RMB); a positive value indicates depreciation of the local currency against the RMB. $\Delta e_{it}^{\$}$ is the corresponding log change against the U.S. dollar; a positive value corresponds to a depreciation against the dollar. Because year fixed effects would prevent separate identification of β_1 and β_2, we follow Boz et al. (2022) and control directly for a set of global factors[3]. Additional controls include the log change in China's consumer price index (ΔCPI_t^{China})

[3] The set of controls include global real GDP growth, the annual change in the global GDP deflator, global export growth, oil price growth rate, and the log of Chicago Board Options Exchange Volatility Index.

and the real GDP growth rate of the importing country ($\Delta RGDP_{i,t}$). All regressions include country-product fixed effects (φ_{ih}) to absorb time-invariant country-product specific factors.

To investigate whether exchange rate pass-through varies with exposure to Chinese lending, we further interact both exchange rates with the lagged loan-to-GDP ratio ($OF_{i,t-1}$, scaled by 100):

$$\Delta p_{iht} = \beta_1 \Delta e_{it}^{RMB} + \beta_2 OF_{i,t-1} + \beta_{rmb}\left(\Delta e_{it}^{RMB} \times OF_{i,t-1}\right) + \beta_{\$}\left(\Delta e_{it}^{\$} \times OF_{i,t-1}\right) + \theta \Delta RGDP_{i,t} + \varphi_{ht} + \varphi_{ih} + \epsilon_{iht} \qquad (2)$$

$OF_{i,t-1}$= 100× $\frac{Loan_{i,t-1}^{USD}}{GDP_{i,t-1}^{USD}}$. Lagging OF by one period mitigates contemporaneous reverse causality between lending and import prices. The product-year fixed effects φ_{ht} absorb all common time-varying product-level shocks (including Chinese inflation and the importer's exchange rate against the dollar), so ΔCPI_t^{China} and $\Delta e_{it}^{\$}$ are no longer separately identified.

3.2 Baseline Results

Table 1 reports the results of exchange rate pass-through into import prices. Columns (1) and (2) correspond to the baseline specification without the lending interaction. A 1% depreciation of the recipient country's currency against the RMB is associated with a 0.65% increase in import prices (column 1). The corresponding coefficient on the dollar exchange rate is 0.232. After controlling for global factors (column 2), the RMB coefficient falls to 0.374 while the dollar coefficient rises to 0.518. Thus, even in bilateral trade between recipient countries and China, the dollar continues to play a dominant role in trade. At the same time, the RMB exchange rate also exerts a statistically significant influence on import prices.

Columns (3)-(4) introduce the interactions with lagged Chinese lending exposure. Higher exposure strengthens RMB exchange rate pass-through and weakens dollar exchange rate pass-through. In the baseline specification with product-year fixed effects (column 4), a 1-percentage-point increase in the loan-to-GDP ratio raises the RMB

exchange rate pass-through coefficient by 0.161 and reduces the dollar exchange rate pass-through coefficient by 0.144.

[Table 1 is about here]

We next replace the dependent variable with the log change in import volume, Δq_{iht}. Table 2 shows the results. Depreciation against either RMB or dollar is expected to reduce import quantities. Column (1) reveals a statistically significant negative coefficient only on the dollar exchange rate, consistent with dollar dominance. When the lending interactions are included (columns 3-4), greater Chinese lending exposure is associated with a more stronger volume response to the RMB exchange rate and a less negative response to the dollar exchange rate. The pattern survives the inclusion of product-year fixed effects.

Taken together, Tables 1 and 2 confirm that the dollar remains important in trade between China and recipient countries, in line with the dollar dominant currency paradigm. However, Chinese official lending shifts the relative importance of the two currencies. It amplifies the role of the RMB and dampens that of the dollar in both price and quantities. We interpret these patterns as evidence that Chinese overseas lending promotes RMB invoicing

[Table 2 is about here]

3.3 Robustness Checks

Appendix Tables A2 and A3 present additional excises. Results remain qualitatively unchanged when we (i) include commodity products, (ii) restrict the sample to 2006-2019 to exclude the COVID-19 period, (iii) cluster standard errors at the country-year level, and (iv) redefine exposure $OF_{i,t-1}= 100\times \frac{Loan_{i,t-1}^{USD}}{GDP_{i,t-2}}$.

4. Mechanism

In the previous section, we establish that even though dollar invoicing is dominant in recipient countries' imports from China, Chinese overseas lending can still promote the RMB invoicing and weakens dollar invoicing. And in this section, we try to explore the mechanism behind Chinese overseas lending. We propose that the effect operates through the commercial procurement channel created by loans (where the recipient

must source from the Chinese suppliers). We prove our hypothesis by showing loan heterogeneity and product heterogeneity analysis.

4.1 Heterogeneity by Lender Type

A natural question is whether the effect of Chinese overseas lending on RMB exchange rate pass-through is uniform across different types of Chinese lenders. AidData classifies Chinese lenders into several categories, including government agencies, state-owned policy banks, state-owned commercial banks, state-owned companies, and state-owned funds. Chinese policy banks (China Development Bank and Export-Import Bank of China) play a central role in financing large-scale infrastructure and strategic projects abroad and typically maintain close relationship with Chinese contractors. In contrast, loans from commercial banks or stated-owned enterprises are more likely to be commercially oriented and less tightly linked to procurement from Chinese suppliers. If the procurement-tying channel is operative, we should therefore observe stronger effects for policy bank loans.

We implement two alternative groupings of lenders. In the first, we combine loans from Chinese government agencies and state-owned policy banks into one category and group all remaining loans (primarily those from commercial banks and stated-owned enterprises) into a second category. In the second, we isolate policy bank loans alone and group all other loans together. We then construct the corresponding loan-to-GDP ratios and interact each with the RMB and dollar exchange rates.

Results are reported in Table 3. Columns (1)-(2) show that only the interaction between government agency plus policy bank loans and the RMB exchange rate is statistically significant and positive (negative) for import prices (quantities). Columns (3)-(4) further isolate the policy bank loans and confirm that only their interactions with the exchange rates remain significant. These findings indicate that loans extended by Chinese policy banks are the primary drivers of stronger RMB exchange rate pass-through, consistent with the commercial procurement channel.

[Table 3 is about here]

4.2 Heterogeneity by Product Type

A second prediction of the procurement-tying channel is that the effect of Chinese overseas lending should be stronger for infrastructure-related products. Most Chinese official loans finance large scale infrastructure projects (roads, power plants, buildings, etc.) that are frequently implemented by Chinese contractors. These projects generate additional demand for related capital goods and materials imported from China, which are more likely to be invoiced in RMB than other products.

To identify infrastructure related products, we use Chinese customs data on exports under the trade mode "goods for overseas contracted projects". The five most frequent HS chapters under this trade mode are 87, 84, 85, 39 and 40. We classify products belonging to these chapters as infrastructure-related and all remaining products as others.

We implement two complementary tests. First, we re-estimate the baseline specification separately for infrastructure-related and other products. Second, we estimate a triple-interaction specification on the full sample that includes the interaction of Chinese loan-to-GDP, the exchange rates, and an infrastructure product dummy. This specification allows us to control for country-year fixed effects, thereby absorbing all time-varying country-level shocks and identifying differential effects across product types within the same country-year.

Results are reported in Table 4. The dependent variable is the log change in import prices in columns (1)-(3) and the log change in import quantities in columns (4)-(6). Columns (1) and (4) present the results for the subsample of infrastructure-related products, while columns (2) and (5) present the results for other products. Columns (3) and (6) report the full sample estimates that include the triple interactions.

The subsample results in columns (1), (2), (4), and (5) show that the interaction between Chinese loan-to-GDP and the RMB exchange rate is substantially larger for infrastructure-related products than for other products, for both import prices and quantities. The full sample estimates in columns (3) and (6) confirm this pattern, the

triple interaction term between Chinese loan-to-GDP, the RMB exchange rate, and an infrastructure product dummy is positive and statistically significant for import prices and negative and statistically significant for import quantities. These findings indicate that the strengthening of RMB exchange rate pass-through is particularly pronounced for infrastructure-related products, consistent with the hypothesis that Chinese overseas loans promote RMB invoicing through the commercial procurement channel associated with infrastructure projects.

[Table 4 is about here]

4.3 Resource-Backed vs. Non-Resource-Backed Loans

We further test the procurement-tying channel by distinguishing resource-backed loans and non-resource-backed loans. When a Chinese policy bank extends a loan that is not secured by commodity revenues, the lender's primary means of risk mitigation is control over the use of proceeds, typically through close ties with Chinese contractors. In contrast, when a loan is secured by commodity exports (such as oil, copper, or cobalt), repayment risk is substantially reduced by contractual pledges and escrow arrangements. In this case, the lender has less incentive to enforce tight procurement linking, which should weaken the effect of RMB invoicing.

To test this prediction, we classify Chinese loans into resource-backed loans (RBL) and non-resource-backed loans (NRBL) and construct the corresponding loan-to-GDP ratios. We first examine the differential effects of the two types of loans on RMB and dollar exchange rate pass-throughs. We then further interact both loan-to-GDP ratios with an infrastructure product dummy and control for country-year fixed effects, allowing us to assess whether the stronger effects of non-resource-backed loans is concentrated in infrastructure-related products.

Results are reported in Table 5. Columns (1)-(2) shows that the interaction between non-resource-backed loan-to-GDP and the RMB exchange rate is positive and statistically significant for import prices (and negative for quantities), while the corresponding interaction for resource-backed loans is insignificant. Columns (3)-(4)

further incorporate product differentiation. The triple interaction involving NRBL is positive and significant for prices and negative for quantities, whereas the corresponding terms for RBL remain insignificant. These results support the hypothesis that non-resource-backed loans, which rely more heavily on procurement typing, are the primary drivers of increased RMB invoicing, especially for infrastructure-related products.

[Table 5 is about here]

5. Other Explanations

In this section, we systematically rule out several alternative explanations for the observed relationship between Chinese overseas lending and stronger RMB ERPT. We show that neither the presence of bilateral RMB swap lines and clearing banks, nor borrower financial distress, political alignment with China, and currency matching between the original denomination of the loans and trade invoicing can explain our findings.

5.1 PBoC Swap Line and RMB Clearing Bank

A potential confounding factor is the network of RMB bilateral swap lines and offshore RMB clearing banks established by the People's Bank of China. Swap lines lower the cost of obtaining RMB liquidity (Bahaj and Reis, 2026) and provide the lender of last resort (Horn et al., 2023) and may themselves encourage RMB invoicing, while the presence of a local RMB clearing bank facilitates RMB settlement. If these institutional arrangements are correlated with Chinese loan exposure, they could bias our estimates.

To address this concern, we use the swap line and RMB clearing bank data from Bahai and Reis (2026) and construct an indicator $SwapLine_{i,t}$ that equals one from the year a country first signs a PBoC swap arrangement onward. We construct an analogous indicator $ClearingBank_{i,t}$ for the presence of an RMB clearing bank. We first include these indicators as additional controls in the baseline regression and then further interact them with the exchange rates.

Results are reported in Table 6. Columns (1)-(3) present estimates of ERPT into

import prices; columns (4)-(6) present the corresponding trade elasticities with respect to exchange rates; and columns (3) and (6) introduce the full set of interactions between these indicators and exchange rates. Across all specifications, the coefficients on interaction terms between Chinese loan-to-GDP exposure and exchange rates remain statistically significant. These findings indicate that the strengthening of RMB exchange rate pass-through associated with Chinese lending is not driven by the presence of swap lines or RMB clearing banks.

[Table 6 is about here]

5.2 Debt Distress

An alternative explanation is that financially distressed borrowers lose bargaining over pricing terms and are therefore more willing to accept RMB invoicing. If this channel is operative, the estimated effects should be concentrated among distressed loans or distressed countries.

We conduct three complementary tests. First, we exploit the Financial_Distress flag in the AidData dataset, which indicates whether a given loan shows evidence that the borrower sought or obtained a rescheduling of terms, or exhibited other signs of repayment difficulty during the loan's repayment period. We classify loans into distressed and non-distressed categories and construct the corresponding loan-to-GDP ratios. We then interact both ratios with the exchange rates. Second, we identify countries that received distress flagged loans in at least five years over 2000-2023, drop these countries from the sample, and re-estimate the baseline specification. Third, we construct a proxy for debt distress, which is Chinese loan exposure relative to the recipient's exports of goods, services, and primary income and include it as an additional control.

Results in Table 7 show that the interaction of non-distressed loan-to-GDP with the RMB exchange rate is statistically significant, whereas the corresponding interaction for distressed loans is not. The baseline effects remain significant after excluding heavily distressed countries and after controlling for the loan-to-export ratio. These findings reject the debt distress hypothesis.

[Table 7 is about here]

5.3 Political Alignment

A second potential mechanism is that countries politically aligned with China engage in more RMB invoicing. If political affinity were the operative channel, the effects of loan exposure should not vary systematically by product type or lender type, which the patterns we already document in the mechanism section. We provide additional evidence by using Chinese official grants as a proxy for political alignment, as grants are more likely to be directed toward politically favored recipient countries.

We construct a grant-to-GDP ratio and re-estimate the baseline specification while controlling for both loan-to-GDP with exchange rates and grant-to-GDP with exchange rates. Results are reported in Table 8. The dependent variable is the log change in import prices in column (1) and the log change in import quantities in column (2). The interactions between loan-to-GDP and exchange rates are still statistically significant, while the interactions between grant-to-GDP and the exchange rates are insignificant. These findings indicate that grant exposure does not moderate RMB exchange rate pass-through and therefore provide further evidence against the political alignment hypothesis.

[Table 8 is about here]

5.4 Currency Matching

A further alternative explanation is that Chinese overseas loans are themselves denominated in RMB, so that recipient countries naturally align the currency composition of their imports with their payment obligations. Under this currency matching hypothesis, the observed strengthening of RMB exchange rate pass-through would simply reflect the original denomination of the loans rather than a broader shift toward RMB invoicing.

To assess this possibility, we first examine the currency composition of Chinese oversea lending. We calculate the annual shares of loan denominated in RMB, in U.S. dollars, and in third currencies (neither RMB nor the dollar), and plot these shares over 2000-2023 in Figure 3. The figure shows that dollar denominated loans consistently

account for more than 60 percent of total lending. In other words, the larger majority of Chinese overseas loans are denominated in U.S. dollar rather than in RMB.

[Figure 3 is about here]

We next classify loans into three categories, RMB denominated, dollar denominated, and other currency denominated, and construct the corresponding loan-to-GDP ratios. We then interact each ratio with the exchange rates and re-estimate the baseline specification.

Results are reported in Table 9. The dependent variable is the log change in import prices in column (1) and the log change in import quantities in column (2). The interaction between dollar denominated loan-to-GDP and the RMB exchange rate is statistically significant (positive for prices and negative for quantities), whereas the corresponding interaction for RMB denominated loans is insignificant. These findings contradict the currency matching hypothesis and show that the strengthening of RMB exchange rate pass-through is driven by dollar denominated rather than RMB denominated loans.

[Table 9 is about here]

6. Placebo Tests

In this section, we conduct a series of placebo tests to assess whether the documented effects are unique to Chinese lending or merely reflect general features of foreign borrowing. First, we replace Chinese loan exposure with World Bank loan exposure and examine its impact on RMB ERPT into imports from China. Second, we test whether World Bank loans (or US official development assistance) moderate dollar ERPT into imports from the US. Third, we examine whether Japanese official development assistance (ODA) moderates yen ERPT into imports from Japan.

6.1 World Bank Loan-to-GDP and ERPTs into Import from China

A natural concern is that any increase in foreign loan exposure, not specifically Chinese lending could affect RMB exchange rate pass-through in recipient countries'

imports from China. To address this concern, we conduct three related exercises using World Bank (IBRD and IDA) loan-to-GDP, sourced from the World Development Indicators.

First, we replace Chinese loan-to-GDP with World Bank loan-to-GDP and re-estimate the baseline specification. Second, we retain Chinese loan exposure and add World Bank loan-to-GDP as an additional control. Third, we jointly include the interactions of both Chinese and World Bank loan exposure with the exchange rates.

Results are reported in Table 10. Columns (1)-(2) show that World Bank loan exposure does not significantly strengthen RMB exchange rate pass-through into import prices from China. Columns (3)-(4) confirm that the Chinese loan interactions remain positive and statistically significant after controlling for World Bank exposure. Columns (5)-(6) further show that when both sets of interactions are included, only the Chinese loan interactions remain significant. These findings indicate that the strengthening of RMB invoicing is specific to Chinese loan exposure and is not dirven by general foreign borrowing.

[Table 10 is about here]

6.2 World Bank Loans and US ODA and Dollar ERPT into Imports from US

To further test uniqueness, we examine whether World Bank loans or US official development assistance (ODA) moderate dollar exchange rate pass-through into imports from the United States. We replace the dependent variables with the log changes in import prices and quantities from the US and interact World Bank loan-to-GDP (or US ODA-to-GDP) with the dollar exchange rate.

Columns (1)-(2) of Table 11 report the baseline dollar exchange rate pass-through. The results show a high degree of dollar pass-through (coefficient of approximately 0.89), consistent with dominant dollar invoicing in trade. Columns (3)-(4) add the interaction between World Bank loan-to-GDP and the dollar exchange rate. Columns (5)-(6) replace World Bank exposure with US ODA-to-GDP and interact it with the dollar exchange rate. Neither the World Bank interaction nor the US ODA interaction

is statistically significant. Thus, World Bank loans or US ODA do not moderate dollar exchange rate pass-through, reinforcing that baseline effects are unique to Chinese lending.

[Table 11 is about here]

6.3 Japanese ODA and Yen EPRT into Import from Japan

Finally, we examine recipient's imports from Japan and test whether Japanese official development assistance moderates yen ERPT. WE expect null effect because Japanese ODA is less tightly tied to bilateral trade and is often intermediated through multilateral procurement systems and Japanese firms exhibit weaker state coordination on invoicing currency.

We estimate an analogous specification, replacing Chinese loan exposure with Japanese ODA-to-GDP. Columns (1)-(2) of Table 12 report the baseline pass-through of the yen and dollar exchange rates into import prices and quantities from Japan. The results show limited yen pass-through and substantial dollar pass-through, indicating that dollar invoicing dominates Japan and recipient trade. Columns (3)-(4) add the interactions of Japanese ODA-to-GDP with both the yen and dollar exchange rates. The interactions of Japanese ODA-to-GDP with both exchange rates are statistically insignificant. These null results further support that the loan effects are specific to Chin

[Table 12 is about here]

7. Conclusion

This paper investigates whether Chinese overseas lending promotes the use of RMB as an invoicing currency in recipient countries' import from China. Combining detailed Chinese overseas lending data from AidData with bilateral trade data, we shows that higher Chinese loan-to-GDP exposure significantly strengthens the pass-through of the RMB bilateral exchange rate into both import prices and import quantities.

We provide evidence on the underlying mechanism. The effects are driven

primarily by loans extended by Chinese policy banks, by non-resource-backed loans, and by infrastructure-related products. These patterns support a procurement-tying channel: Chinese official loans raise demand for infrastructure-related products that are more likely to be invoiced in RMB.

We rule out several alternative explanations. The results are not driven by the presence of bilateral currency swap lines, by financially distressed borrowers, by political alignment with China, or by the original currency denomination of the loans. Placebo testes further reinforce the uniqueness of the findings: World Bank loan exposure does not affect RMB exchange rate pass-through into recipient's imports from China, nor does it moderate dollar exchange rate pass-through into imports from the US and Japanese official development assistance does not raise yen exchanger rate pass-through into imports from Japan.

Our findings identify a lending-by-invoicing channel through which China's rise as the world's largest official creditor may accelerate the internationalization of the RMB. By linking sovereign lending to trade invoicing, the paper contributes to the literature on currency choice influenced by Chinese overseas lending. Future research can explore the potential spillovers to invoicing patters in third country trade.

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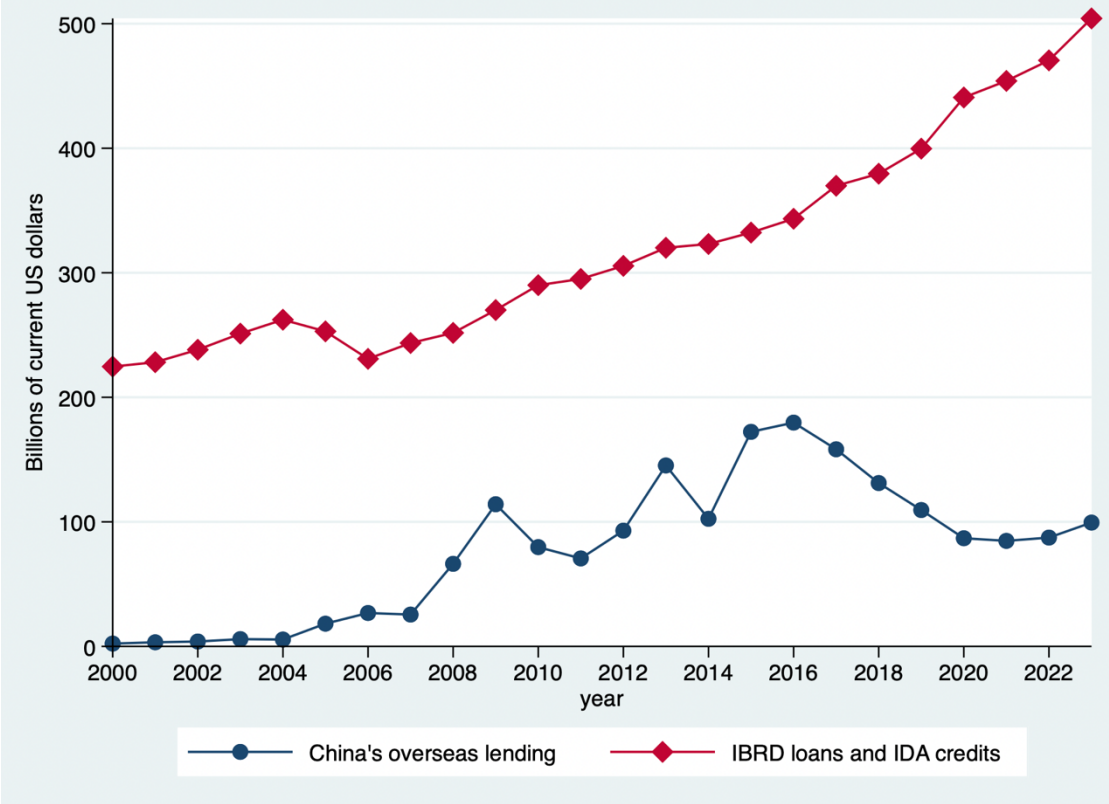
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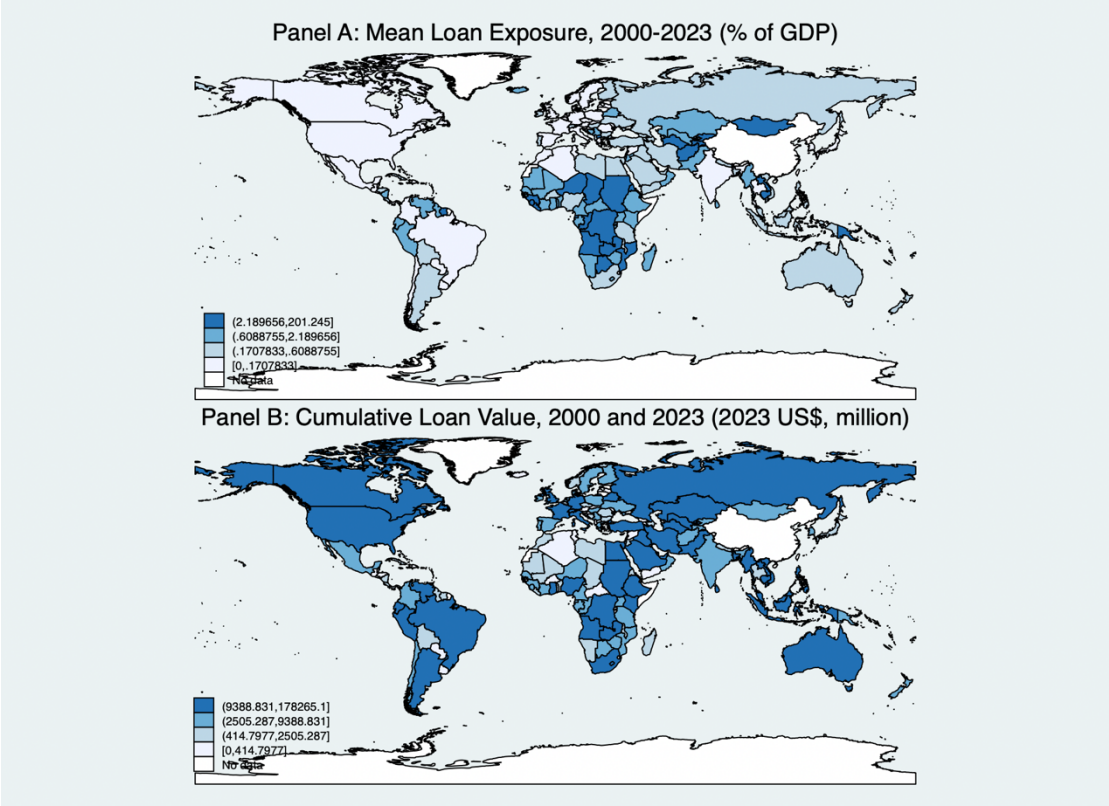
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Figure 1: China and World Bank Annual Overseas Lending, 2000-2023



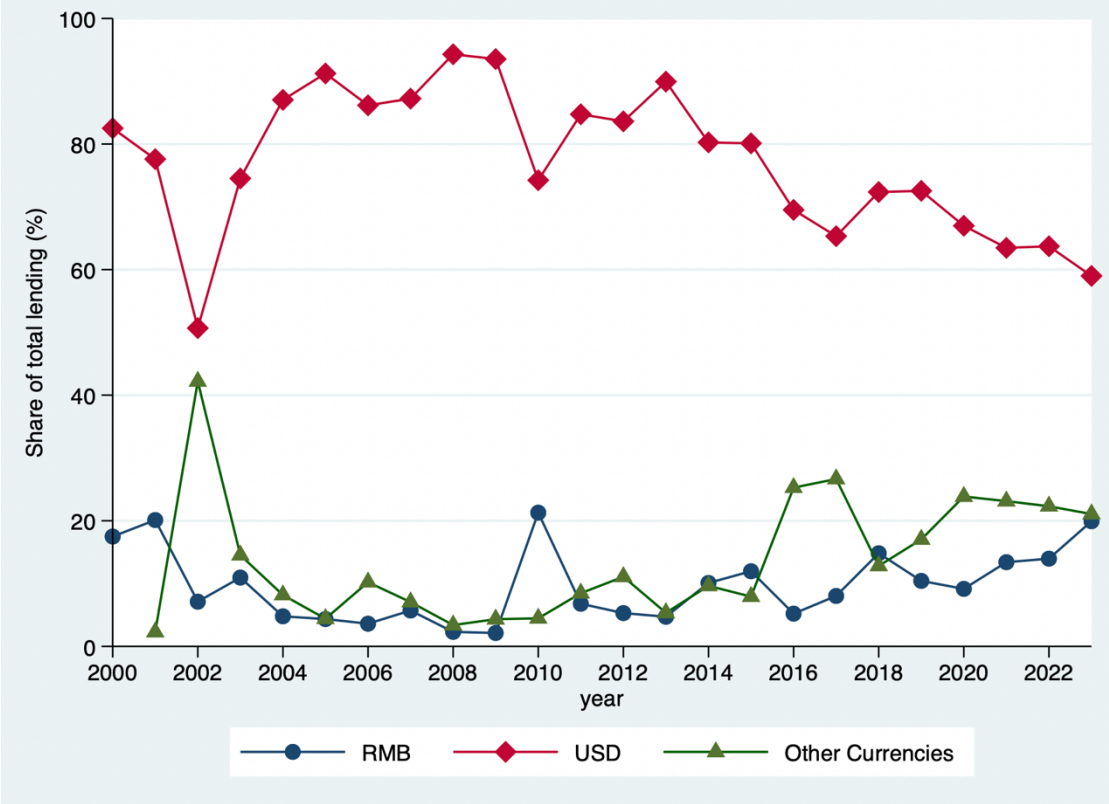
Notes: This figure plots the annual value of overseas lending extended by China and IBRD&IDA (World Bank), measured in billions of nominal US dollars.

Figure 2: Geographical Distribution of Chinese Overseas Lending, 2000-2023



Note: Panel A reports each recipient country's mean annual loan-to-GDP ratio (%). Panel B reports the cumulative loan value in constant 2023 US dollars. Darker shades indicate higher values.

Figure 3: Currency Composition of Chinese Overseas Lending, 2000-2023



Notes: This figure plots the share of Chinese overseas lending in RMB, dollar and other currencies, measured in percent of the total value of lending in that year. Other currencies mean aggregated all currencies other than the RMB and the US dollar.

Table 1: Exchange Rate Pass-Through into Import Prices

	The dependent variable is Δp_{ijt}			
	(1)	(2)	(3)	(4)
$\Delta e_{i,RMB,t}$	0.650***	0.374***	0.286**	0.873***
	(6.201)	(3.013)	(2.185)	(25.272)
$\Delta e_{i,RMB,t}$ ×OF			0.179***	0.161**
			(2.736)	(2.316)
$\Delta e_{i,\$,t}$	0.232**	0.518***	0.596***	
	(2.188)	(4.107)	(4.453)	
$\Delta e_{i,\$,t}$ ×OF			-0.161**	-0.144**
			(-2.610)	(-2.192)
OF			-0.000	0.000
			(-0.112)	(0.052)
Global controls	No	Yes	Yes	Yes
Country×Product FE	Yes	Yes	Yes	Yes
Product×Year FE	No	No	No	Yes
Observations	2,425,394	2,425,394	2,425,394	2,423,133
R^2	0.079	0.079	0.080	0.111
Number of countries	116	116	116	116

Notes: The dependent variable is the log change in unit values of imports from China. All regressions include a constant term, China's CPI inflation, and the importer's real GDP growth rate. Standard errors are clustered at the country level; t-statistics are reported in parentheses. *, **, *** indicate the significance level of 10%, 5%, and 1%, respectively.

Table 2: Elasticity of Import Volumes with Respect to Exchange Rate

	The dependent variable is Δq_{ijt}			
	(1)	(2)	(3)	(4)
$\Delta e_{i,RMB,t}$	0.179	0.012	0.141	-0.194*
	(1.061)	(0.066)	(0.742)	(-1.866)
$\Delta e_{i,RMB,t}$ ×OF			-0.301***	-0.247***
			(-3.492)	(-2.772)
$\Delta e_{i,\$,t}$	-0.501***	-0.302	-0.360	
	(-2.691)	(-1.361)	(-1.629)	
$\Delta e_{i,\$,t}$ ×OF			0.219***	0.162*
			(2.635)	(1.914)
OF			0.004	0.007**
			(1.226)	(2.082)
Global controls	No	Yes	Yes	Yes
Country×Product FE	Yes	Yes	Yes	Yes
Product×Year FE	No	No	No	Yes
Observations	2,437,162	2,437,162	2,437,162	2,434,901
R^2	0.069	0.070	0.071	0.104
Number of countries	116	116	116	116

Notes: The dependent variable is the log change in import quantities from China. All regressions include a constant term, China's CPI inflation, and the importer's real GDP growth rate. Standard errors are clustered at the country level; t-statistics are reported in parentheses. *, **, *** indicate the significance level of 10%, 5%, and 1%, respectively.

Table 3: Loan Heterogeneity by Lender Types

	Govern.+Policy Banks vs. others		Policy Banks vs. Others	
	Δp_{ijt}	Δq_{ijt}	Δp_{ijt}	Δq_{ijt}
	(1)	(2)	(3)	(4)
$\Delta e_{it}^{RMB} \times OF_{policy}$	0.221**	-0.297**	0.208*	-0.327**
	(2.202)	(-2.388)	(1.868)	(-2.250)
$\Delta e_{it}^{RMB} \times OF_{commerical}$	0.023	-0.187	0.078	-0.096
	(0.123)	(-0.696)	(0.419)	(-0.365)
$\Delta e_{it}^{\$} \times OF_{policy}$	-0.194**	0.197*	-0.185*	0.230*
	(-2.102)	(1.726)	(-1.829)	(1.765)
$\Delta e_{it}^{\$} \times OF_{commerical}$	-0.053	0.221	-0.092	0.059
	(-0.278)	(0.742)	(-0.527)	(0.225)
Country×Product FE	Yes	Yes	Yes	Yes
Product×Year FE	Yes	Yes	Yes	Yes
Observations	2,423,133	2,434,901	2,423,133	2,434,901
R^2	0.111	0.104	0.111	0.104
Number of countries	116	116	116	116

Notes: This table examines the heterogeneous effects of Chinese overseas loans by lender types. The dependent variable is the log change in import prices in columns (1) and (3) and the log change in import quantities in columns (2) and (4). In the first two columns, loans are grouped into those extended by Chinese government agencies and state-owned policy banks versus all remaining lenders. In columns (3) and (4), loans are grouped into those extended by state-owned policy banks alone versus all other lenders. All regressions control for the respective loan-to-GDP ratios and exchange rates, and include a constant term, country-product and product-year fixed effects, and the importer's real GDP growth rate. Standard errors are clustered at the country level, and the corresponding t-statistics are provided in parentheses. *, **, *** indicate the significance level of 10%, 5%, and 1%, respectively.

Table 4: Heterogeneity by Product Type

	Δp_{ijt}			Δq_{ijt}		
	Infra. products	Other products	Full sample	Infra. products	Other products	Full sample
	(1)	(2)	(3)	(4)		
Δe_{it}^{RMB} ×OF	0.217**	0.135**		-0.309***	-0.214**	
	(2.402)	(2.037)		(-2.915)	(-2.289)	
Δe_{it}^{RMB} ×OF×Infra			0.123**			-0.150*
			(2.080)			(-1.746)
$\Delta e_{it}^{\$}$ ×OF	-0.195**	-0.120*		0.203**	0.141	
	(-2.299)	(-1.927)		(1.992)	(1.598)	
$\Delta e_{it}^{\$}$ ×OF×Infra			-0.113**			0.116
			(-2.087)			(1.424)
Country×Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Product×Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Country×Year FE	No	No	Yes	No	No	Yes
Observations	711,843	1,711,290	2,423,128	727,228	1,707,673	2,434,895
R^2	0.110	0.111	0.137	0.099	0.106	0.126
Number of countries	116	116	116	116	116	116

Notes: This table examines the heterogeneous effects of Chinese overseas loans by product type. The dependent variable is the log change in import prices in columns (1)-(3) and the log change in import quantities in columns (4)-(6). Columns (1) and (4) report results for the subsample of infrastructure related products, columns (2) and (5) for non-infrastructure products, and columns (3) and (6) for the full sample that includes the triple interaction between Chinese loan-to-GDP, exchange rates and infrastructure dummy. All regressions control for the loan-to-GDP ratio and exchange rates, and include a constant, country-product and product-year fixed effects, and the importer's real GDP growth rate. Columns (3) and (6) additional include country-year fixed effects. Standard errors are clustered at the country level, and the corresponding t-statistics are provided in parentheses. *, **, *** indicate the significance level of 10%, 5%, and 1%, respectively.

Table 5: Resource-Backed vs Non-Resource-Backed Loans

	RBL vs. NRBL		Triple interaction with Infra	
	Δp_{ijt}	Δq_{ijt}	Δp_{ijt}	Δq_{ijt}
	(1)	(2)	(3)	(4)
$\Delta e_{it}^{RMB} \times OF_{RBL}$	-0.047	-0.365		
	(-0.185)	(-1.016)		
$\Delta e_{it}^{RMB} \times OF_{NRBL}$	0.187***	-0.259***		
	(3.069)	(-2.964)		
$\Delta e_{it}^{\$} \times OF_{RBL}$	0.043	0.248		
	(0.185)	(0.784)		
$\Delta e_{it}^{\$} \times OF_{NRBL}$	-0.185***	0.200**		
	(-2.867)	(2.072)		
$\Delta e_{it}^{RMB} \times OF_{RBL}$ ×Infra			0.180	-0.413
			(0.902)	(-1.029)
$\Delta e_{it}^{RMB} \times OF_{NRBL}$ ×Infra			0.115*	-0.101
			(1.809)	(-1.154)
$\Delta e_{it}^{\$} \times OF_{RBL}$ ×Infra			-0.166	0.364
			(-0.918)	(1.017)
$\Delta e_{it}^{\$} \times OF_{NRBL}$ ×Infra			-0.103*	0.035
			(-1.669)	(0.379)
Country×Product FE	Yes	Yes	Yes	Yes
Product×Year FE	Yes	Yes	Yes	Yes
Country×Year FE	No	No	Yes	Yes
Observations	2,423,133	2,434,901	2,423,128	2,434,895
R^2	0.111	0.104	0.137	0.126
Number of countries	116	116	116	116

Notes: This table examines the heterogenous effects of Chinese overseas loans by distinguishing resource-backed-loans (RBL) and non-resource-backed loans (NRBL). The dependent variable is the log change in import prices in columns (1) and (3), and the log change in import quantities in columns (2) and (4). Columns (1)-(2) report the differential effects of resource-backed and non-resource-backed loan-to-GDP ratios interacted with the exchange rates. Columns (3)-(4) further include triple interactions with an infrastructure product dummy and control for country-year fixed effects. All regressions control for the respective loan-to-GDP ratios and exchange rates, and include a constant, country-product and product-year fixed effects, and the importer's real GDP growth rate. Standard errors are clustered at the country level, and the corresponding t-statistics are provided in parentheses. *, **, *** indicate the significance level of 10%, 5%, and 1%, respectively.

Table 6: PBoC Swap Line and RMB Clearing Bank

	Δp_{ijt}			Δq_{ijt}		
	(1)	(2)	(3)	(4)	(5)	(6)
Δe_{it}^{RMB}	0.882***	0.882***	0.816***	-0.203**	-0.204**	-0.354***
	(23.940)	(23.878)	(12.763)	(-2.020)	(-2.012)	(-2.719)
OF	0.000	0.000	0.000	0.007**	0.007**	0.007*
	(0.092)	(0.088)	(0.046)	(2.057)	(2.027)	(1.891)
Δe_{it}^{RMB} ×OF	0.163**	0.162**	0.151**	-0.248***	-0.249***	-0.231**
	(2.334)	(2.330)	(2.193)	(-2.788)	(-2.792)	(-2.599)
$\Delta e_{it}^{\$}$ ×OF	-0.146**	-0.146**	-0.133**	0.164*	0.165*	0.155*
	(-2.217)	(-2.212)	(-2.024)	(1.935)	(1.942)	(1.845)
SwapLine	-0.043***	-0.043***	-0.053***	0.045	0.044	0.027
	(-2.785)	(-2.798)	(-3.118)	(1.578)	(1.544)	(0.889)
SwapLine× Δe_{it}^{RMB}			0.222			0.000
			(0.871)			(0.000)
ClearingBank		0.002	0.001		0.019	0.033*
		(0.212)	(0.107)		(1.048)	(1.872)
ClearingBank× Δe_{it}^{RMB}			-0.562**			0.974**
			(-2.087)			(2.396)
Country×Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Product×Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,423,133	2,423,133	2,423,133	2,434,901	2,434,901	2,434,901
R^2	0.111	0.111	0.111	0.104	0.104	0.104
Number of countries	116	116	116	116	116	116

Notes: This table reports results that control for PBoC swap lines and RMB clearing banks. The dependent variable in columns (1)-(3) is log change in import price, and in columns (4)-(6) it is log change in import quantities. Columns (1) and (4) include the swap line indicator; columns (2) and (5) additionally include the RMB clearing bank variable. Columns (3) and (6) further add the interaction terms between the swap line and clearing bank indicators and the RMB and dollar exchange rates. To conserve space, only the interactions with the RMB exchange rate are reported. All regressions include a constant term, country-product and product-year fixed effects, and control for the importer's real GDP growth rate. Standard errors are clustered at the country level, and the corresponding t-statistics are provided in parentheses. *, **, *** indicate the significance level of 10%, 5%, and 1%, respectively.

Table 7: Debt Distress

	Loan heterogeneity		Drop distressed countries		Control debt distress	
	Δp_{ijt}	Δq_{ijt}	Δp_{ijt}	Δq_{ijt}	Δp_{ijt}	Δq_{ijt}
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta e_{it}^{RMB} \times OF_{distressed}$	0.075	-0.208				
	(0.850)	(-1.531)				
$\Delta e_{it}^{RMB} \times OF_{others}$	0.253**	-0.303**				
	(2.540)	(-1.985)				
$\Delta e_{it}^{\$} \times OF_{distressed}$	-0.065	0.119				
	(-0.774)	(0.925)				
$\Delta e_{it}^{\$} \times OF_{Others}$	-0.259***	0.266*				
	(-2.658)	(1.783)				
Δe_{it}^{RMB} ×OF			0.200**	-0.258	0.157**	-0.240***
			(2.081)	(-1.655)	(2.354)	(-2.659)
$\Delta e_{it}^{\$}$ ×OF			-0.166**	0.263*	-0.140**	0.156*
			(-2.036)	(1.919)	(-2.215)	(1.819)
Loan-to-exports					0.001	-0.001
					(0.843)	(-0.357)
Country×Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Product×Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,423,133	2,434,901	1,728,277	1,736,288	2,335,344	2,346,997
R^2	0.111	0.104	0.110	0.112	0.112	0.105
Number of countries	116	116	83	83	112	112

Notes: This table presents the results of debt distress analysis. The dependent variable is log change of import prices in columns (1), (3), and (5), and the log change in import quantities in columns (2), (4), and (6). Columns (1)-(2) separately include the interactions of distressed and non-distressed loan-to-GDP ratios with the RMB and dollar exchange rates. Columns (3)-(4) drop countries that received distress-flagged loans in at least five years over 2000-2023. Columns (5)-(6) additionally control for the loan-to-exports ratio as a proxy for debt distress. All regressions include a constant term, country-product and product-year fixed effects, and control for the importer's real GDP growth rate. Standard errors are clustered at the country level, and the corresponding t-statistics are provided in parentheses. *, **, *** indicate the significance level of 10%, 5%, and 1%, respectively.

Table 8: Political Alignment

	Δp_{ijt}	Δq_{ijt}
	(1)	(2)
$\Delta e_{it}^{RMB} \times OF_{grant}$	-0.043	0.342
	(-0.138)	(0.812)
$\Delta e_{it}^{RMB} \times OF_{loan}$	0.165**	-0.253***
	(2.333)	(-2.851)
$\Delta e_{it}^{\$} \times OF_{grant}$	0.339	-0.630
	(0.704)	(-1.022)
$\Delta e_{it}^{\$} \times OF_{loan}$	-0.147**	0.166*
	(-2.194)	(1.976)
Country×Product FE	Yes	Yes
Product×Year FE	Yes	Yes
Observations	2,423,133	2,434,901
R^2	0.111	0.104
Number of countries	116	116

Note: This table reports the results of politically alignment analysis. The dependent variable is the log change in import price in column (1) and log change in import quantities in column (2). All regressions control for the exchange rates, the loan-to-GDP ratio, and the grant-to-GDP ratio; only the coefficients on the interaction terms are reported to conserve space. All regressions include a constant term, country-product and product-year fixed effects, and control for the importer's real GDP growth rate. Standard errors are clustered at the country level, and the corresponding t-statistics are provided in parentheses. *, **, *** indicate the significance level of 10%, 5%, and 1%, respectively.

Table 9: Currency Matching

	Δp_{ijt}	Δq_{ijt}
	(1)	(2)
$\Delta e_{it}^{RMB} \times OF_{RMB}$	0.347	-0.541*
	(1.527)	(-1.668)
$\Delta e_{it}^{RMB} \times OF_{\$}$	0.142**	-0.182*
	(1.988)	(-1.728)
$\Delta e_{it}^{RMB} \times OF_{other}$	0.022	0.177
	(0.075)	(0.512)
$\Delta e_{it}^{\$} \times OF_{RMB}$	-0.271	0.369
	(-1.320)	(1.320)
$\Delta e_{it}^{\$} \times OF_{\$}$	-0.128*	0.103
	(-1.945)	(1.038)
$\Delta e_{it}^{\$} \times OF_{other}$	-0.031	0.051
	(-0.118)	(0.228)
Country×Product FE	Yes	Yes
Product×Year FE	Yes	Yes
Observations	2,423,133	2,434,901
R^2	0.111	0.104
Number of countries	116	116

Note: This table reports the results of currency matching analysis. The dependent variable is the log change in import price in column (1) and log change in import quantities in column (2). All regressions control for the exchange rates, the currency loan-to-GDP ratios; only the coefficients on the interaction terms are reported to conserve space. All regressions include a constant term, country-product and product-year fixed effects, and control for the importer's real GDP growth rate. Standard errors are clustered at the country level, and the corresponding t-statistics are provided in parentheses. *, **, *** indicate the significance level of 10%, 5%, and 1%, respectively.

Table 10: World Bank's Loans and ERPT into Imports from China

	Replace Chinese loan-to-GDP with WB loan-to-GDP		Add WB loan-to-GDP in the baseline regression		Control WB loan-to-GDP×ERs, Chinese loan-to-GDP×ERs	
	Δp_{ijt}	Δq_{ijt}	Δp_{ijt}	Δq_{ijt}	Δp_{ijt}	Δq_{ijt}
	(1)	(2)	(3)	(4)	(5)	(6)
Δe_{it}^{RMB} ×WB loan-to-GDP	0.023	-0.058**			0.021	-0.057**
	(1.395)	(-2.119)			(1.241)	(-2.119)
$\Delta e_{it}^{\$}$ ×WB loan-to-GDP	-0.020	0.030			-0.016	0.021
	(-1.256)	(0.940)			(-0.995)	(0.711)
Δe_{it}^{RMB} ×OF			0.168**	-0.250***	0.166**	-0.252***
			(2.406)	(-2.779)	(2.386)	(-2.803)
$\Delta e_{it}^{\$}$ ×OF			-0.151**	0.165*	-0.148**	0.164*
			(-2.278)	(1.930)	(-2.256)	(1.921)
Country×Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Product×Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,423,133	2,434,901	2,423,133	2,434,901	2,423,133	2,434,901
R^2	0.111	0.104	0.111	0.104	0.111	0.104
Number of countries	116	116	116	116	116	116

Notes: This table examine whether World Bank (WB) lending to the importer affects ERPT in the importer's imports from China. Columns (1) and (2) replace Chinese loans/GDP with the importer's WB loans/GDP; columns (3) and (4) re-estimate the baseline specification with the importer's WB loans/GDP added as an addition control; and columns (5) and (6) include the interactions of both WB loans/GDP and Chinese loans/GDP with the exchange rates. Only interaction coefficients are reported. All regressions include a constant term, country-product and product-year fixed effects, and control for the importer's real GDP growth rate. Standard errors are clustered at the country level, and the corresponding t-statistics are provided in parentheses. *, **, *** indicate the significance level of 10%, 5%, and 1%, respectively.

Table 11: World Bank's Loans and ERPT into Imports from US

	ER only		ER×WB loan-to-GDP		ER×US ODA-to-GDP	
	Δp_{ijt}	Δq_{ijt}	Δp_{ijt}	Δq_{ijt}	Δp_{ijt}	Δq_{ijt}
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta e_{it}^{\$}$	0.890***	-0.322***	0.890***	-0.312***	0.900***	-0.399***
	(40.518)	(-3.878)	(28.362)	(-3.578)	(32.370)	(-6.469)
WB loan-to-GDP			-0.002**	0.002*		
			(-2.242)	(1.750)		
$\Delta e_{it}^{\$}$ ×WB loan-to-GDP			-0.001	-0.002		
			(-0.175)	(-0.176)		
US ODA-to-GDP					-0.006	0.005
					(-1.200)	(0.596)
$\Delta e_{it}^{\$}$ ×US ODA-to-GDP					-0.030	0.219***
					(-1.030)	(2.851)
Country×Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Product×Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,142,369	2,152,886	2,142,369	2,152,886	2,142,369	2,152,886
R^2	0.103	0.098	0.103	0.098	0.103	0.098
Number of countries	118	118	118	118	118	118

Notes: This table presents the results of placebo test. The dependent variables are log change of importer's import prices (columns (1), (3), (5)) and quantities (columns (2), (4), and (6)) from the US. Columns (1)-(2) report the dollar ERPT into import prices and quantities; columns (3)-(4) add the interaction between dollar exchange rate and WB loan-to-GDP; and columns (5)-(6) include the interaction term between dollar exchange rate and ODA received from the US, scaled by GDP. The sample period is 2000-2023; the results are unchanged when the sample is restricted to the baseline period 2006-2023. All regressions include a constant term, country-product and product-year fixed effects, and control for the importer's real GDP growth rate. Standard errors are clustered at the country level, and the corresponding t-statistics are provided in parentheses. *, **, *** indicate the significance level of 10%, 5%, and 1%, respectively.

Table 12: Japan's ODA-to-GDP and ERPT into Imports from Japan

	ER only		ER× Japan ODA-to-GDP	
	Δp_{ijt}	Δq_{ijt}	Δp_{ijt}	Δq_{ijt}
	(1)	(2)	(3)	(4)
Δe_{it}^{Japan}	0.095	0.235**	0.865***	-0.377***
	(1.399)	(2.612)	(18.950)	(-4.538)
$\Delta e_{it}^{\$}$	0.748***	-0.608***		
	(10.406)	(-5.911)		
Japan ODA-to-GDP			-0.010	0.003
			(-0.865)	(0.176)
Δe_{it}^{Japan} ×Japan ODA-to-GDP			-0.091	0.029
			(-1.456)	(0.187)
$\Delta e_{it}^{\$}$ ×Japan ODA-to-GDP			-0.082	-0.045
			(-0.697)	(-0.261)
Global Controls	Yes	Yes	No	No
Country×Product FE	Yes	Yes	Yes	Yes
Product×Year FE	No	No	Yes	Yes
Observations	1,189,677	1,194,802	1,185,619	1,190,749
R^2	0.068	0.066	0.132	0.132
Number of countries	118	118	118	118

Notes: This table examine how Japanese official development assistance (ODA) affects exchange rate pass-through (ERPT) into the importer's imports from Japan. The dependent variable is the log change in the importer's import price in columns (1) and (3), and the log change in import quantity in columns (2) and (4). Columns (1)-(2) report the pass-through of the bilateral Japan importer exchange rate into import prices and quantities; columns (3) and (4) add its interaction with Japan ODA-to-GDP. The sample period is 2000-2023; the results are unchanged when the sample is restricted to the baseline period 2006-2023. All regressions include a constant term, country-product and product-year fixed effects, and control for the importer's real GDP growth rate. Standard errors are clustered at the country level, and the corresponding t-statistics are provided in parentheses. *, **, *** indicate the significance level of 10%, 5%, and 1%, respectively.

Online Appendix

Appendix A

Comtrade

UN Comtrade provides detailed annual customs data for a large set of countries at HS 6-digit product level with information about destination country, USD value, quantity, and units of imports and exports. We keep non-commodity goods in our main sample, except noted otherwise. We follow Gopinath et al. (2020) and define commodities broadly as HS chapters 1-27 and 72-83. We exclusively use Comtrade data reported by the importing country as importer-reported data is regard as being more reliable since imports generate tariff revenues. (Gopinath et al., 2020).

World Development Indicator Data

The exchange rate is World Bank's "alternative conversion factor" series (PA.NUS.ATLS), which corrects for redenominations currency substitution, and is measured as annual average of daily rates. Consumer prices are given by FP.CPT.TOTL. Real GDP in USD dollars are given by "NY.GDP.MKTP. KD". Nominal GDP in USD dollars are given by "NY.GDP.MKTP.CD". GDP deflator is the ratio of nominal GDP to real GDP.

FRED data

Crude oil price is source from https://fred.stlouisfed.org/series/DCOILWTICO. We then deflate the oil price with world GDP deflator and then calculate the oil price growth rate.

CBOE volatility index (VIX) is sourced from https://fred.stlouisfed.org/series/VIXCLS#

Country groups.

Dollarized and dollar-pegged countries are Antigua and Barbuda, Aruba, Behamas, Bahrain, Barbados, Belize, Bermuda, Cayman Islands, Djibouti, Dominica, Ecuador, El Salvador, FS Micronesia, Grenada, Jordan, Lebanon, Oman, Palau, Qatar, Saint Kitts and Nevis, Saint and Lucia, Saint Vincent and the Grenadines, Saudi Arabia, Timor-Leste, Turks and Caicos Islands, USA, United Arab Emirates.

ODA ineligible countries are Andorra, Aruba, Australia, Austria, Bahamas, Belgium, Bermuda, Brunei Darussalam, Bulgaria, Canada, Cayman Islands, Cyprus, Czechia, Denmark, Estonia, Faeroe Islands, Finland, France, French Polynesia, Germany, Greece, Greenland, Hungary, Iceland, Ireland, Israel, Italy, Japan, Kuwait, Latvia, Lithuania, Luxembourg, Netherlands, New Caledonia, New Zealand, Norway, Poland, Portugal, Qatar, Korea, Romania, Russian Federation, Singapore, Slovakia, Spain, Sweden, Switzerland, USA, United Arab Emirates, United Kingdom

Appendix Tables

TableA1: Summary Statistics

VarName	N	Mean	Median	SD	P5	Min	P95	Max
Panel A: Country-Product-Year level								
Δp_{ijt}	3,156,597	0.060	0.052	0.885	-1.480	-3.625	1.590	4.197
Δq_{ijt}	3,162,508	0.113	0.085	1.537	-2.518	-6.188	2.815	6.175
Panel B: Country-Year level								
Loan-to-GDP%	885	1.521	0.498	2.814	0.007	0.000	6.732	23.585
Δe_{it}^{RMB}	1,714	0.053	0.030	0.135	-0.069	-0.336	0.235	2.291
$\Delta e_{it}^{\$}$	1,714	0.046	0.017	0.137	-0.081	-0.292	0.223	2.341
Importer's real GDP growth rate	1,711	0.035	0.039	0.057	-0.045	-0.784	0.092	0.591

Notes: This table reports the summary statistics of the main variables used in the regression. The sample excludes dollar-pegged and ODA-ineligible countries as well as commodity products and covers the period 2006-2023.

这里 loan/GDP 是原始的比例，哪些没有收到 loan 的国家其 loan 就为 missing。但是我们在回归的时候，没有 loan 的国家我们替换为 0.这里这样做的原因是，如果现在就替换为 0，loan/GDP 的 median 就为 0 了，零值太多了。我们报告了 maxmin 值后还报告了 p5，p95，是因为我感觉这个 maxmin 值还是有点大，我们可不可以只报告 p95，p5 值。我看 Li et al 2015 这篇文章也是报告的 p10，p90 值，我觉得这篇文章的缩尾方式，uv 的极端值应该也很大。我也试了这篇文章的缩尾，winsorze 上下 5%，by(HS2 year),我们的结果也是显著的，只不过最大最小值比我们 winsroze 上下 1%, by(year)还大。

Table A2: Exchange Rate Pass-Through into Prices-robustness checks

	The dependent variable is Δp_{ijt}			
	Include all products	Restrict to 2006-2019	Cluster at country year	OF=100*$\frac{Loan_{i,t-1}^{USD}}{GDP_{i,t-2}}$
	(1)	(2)	(3)	(4)
Δe_{it}^{RMB}	0.876***	0.808***	0.873***	0.888***
	(27.442)	(14.940)	(25.914)	(23.598)
OF	0.000	0.001	0.000	-0.002
	(0.078)	(0.197)	(0.059)	(-0.577)
Δe_{it}^{RMB} ×OF	0.161**	0.166**	0.161**	0.153**
	(2.428)	(2.176)	(2.029)	(2.555)
$\Delta e_{it}^{\$}$ ×OF	-0.145**	-0.144**	-0.144*	-0.132**
	(-2.300)	(-2.027)	(-1.945)	(-2.465)
Country×Product FE	Yes	Yes	Yes	Yes
Product×Year FE	Yes	Yes	Yes	Yes
Observations	3,107,011	1,811,613	2,423,133	2,150,200
R^2	0.113	0.115	0.111	0.111
Number of countries	116	115	116	115

Notes: This table reports robustness checks for the baseline specification reported in column (4) of Table 1. Column (1) extends the estimation sample to include commodity products; column (2) restricts the sample period to 2006-2019; column (3) clusters standard errors at the country-year level; column (4) redefines importer's exposure to Chinese loan as OF = 100× $\frac{Loan_{i,t-1}^{USD}}{GDP_{i,t-2}^{USD}}$, lagging the denominator by one additional to further mitigate reverse causality. All regressions include a constant term, control for the importer's real GDP growth rate, and include the country-product and product-year fixed effects. Standard errors are clustered at the country level except in column (3), and the corresponding t-statistics are reported in parentheses. *, **, *** indicate the significance level of 10%, 5%, and 1%, respectively.

Table A3: Trade Elasticity with Respect to Exchange Rate-robustness checks

	The dependent variable is Δq_{ijt}			
	Include all products	Restrict to 2006-2019	Cluster at country year	OF=100*$\frac{Loan_{i,t-1}^{USD}}{GDP_{i,t-2}}$
	(1)	(2)	(3)	(4)
Δe_{it}^{RMB}	-0.195*	-0.279**	-0.194***	-0.187*
	(-1.763)	(-2.508)	(-2.888)	(-1.770)
OF	0.007**	0.007*	0.007**	0.010***
	(2.227)	(1.869)	(1.994)	(2.823)
Δe_{it}^{RMB} ×OF	-0.234***	-0.238**	-0.247**	-0.215**
	(-2.676)	(-2.412)	(-2.364)	(-2.240)
$\Delta e_{it}^{\$}$ ×OF	0.154*	0.160*	0.162*	0.110
	(1.862)	(1.723)	(1.676)	(1.210)
Country×Product FE	Yes	Yes	Yes	Yes
Product×Year FE	Yes	Yes	Yes	Yes
Observations	3,112,644	1,820,874	2,434,901	2,158,827
R^2	0.108	0.116	0.104	0.105
Number of countries	116	115	116	115

Notes: This table reports robustness checks for the baseline specification reported in column (4) of Table 1. Column (1) extends the estimation sample to include commodity products; column (2) restricts the sample period to 2006-2019; column (3) clusters standard errors at the country-year level; column (4) redefines importer's exposure to Chinese loan as OF = 100× $\frac{Loan_{i,t-1}^{USD}}{GDP_{i,t-2}^{USD}}$, lagging the denominator by one additional to further mitigate reverse causality. All regressions include a constant term, control for the importer's real GDP growth rate, and include the country-product and product-year fixed effects. Standard errors are clustered at the country level except in column (3), and the corresponding t-statistics are reported in parentheses. *, **, *** indicate the significance level of 10%, 5%, and 1%, respectively.